Introduction to Optimal Control-Chapter 3

Dept. of Automatic Control
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Table of Contents

Part II. Dynamic Programming

1. Bellman's Principle of Optimality
2. DP in Discrete-Time Systems
3. DP in Continuous-Time Systems
4. ADP (Numerical Methods)
5. Examples (Course Project)

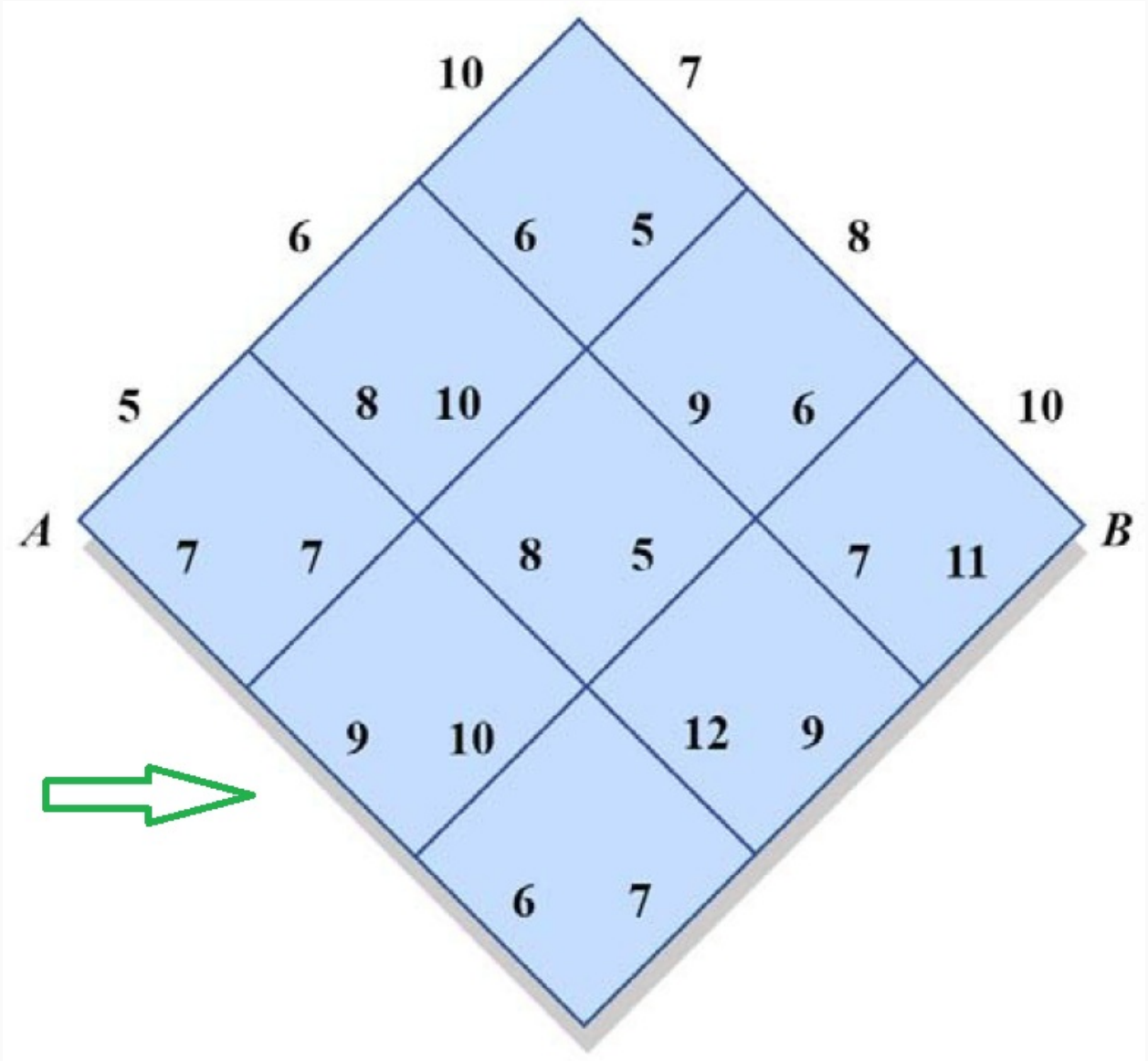
Life can only be understood going backwards, but it must be lived going forwards. Kierkegaard

The slides will be updated frequently. Do not print.

Bellman's Principle of Optimality

A Shortest Path Problem

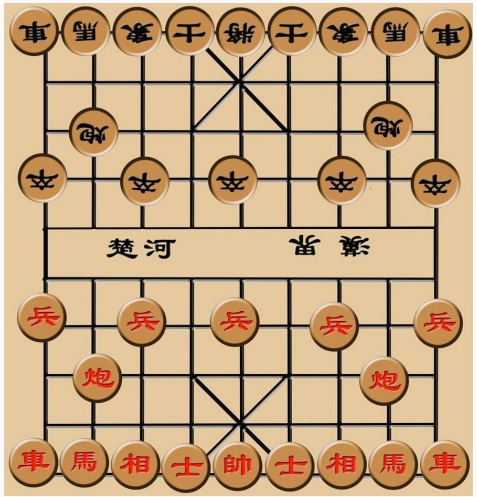
Classic robot navigation and/or aircraft flight path problems

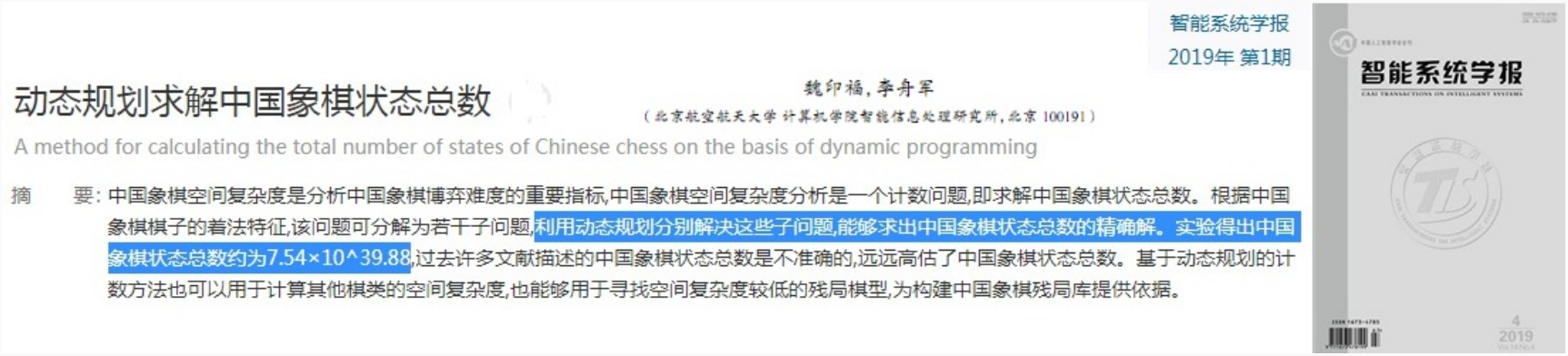


More Examples on Multi-Stage Decision Problems









智能系统学报
2019年 第1期

动态规划求解中国象棋状态总数

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A method for calculating the total number of states of Chinese chess on the basis of dynamic programming

摘　要：中国象棋空间复杂度是分析中国象棋博弈难度的重要指标，中国象棋空间复杂度分析是一个计数问题，即求解中国象棋状态总数。根据中国象棋棋子的着法特征，该问题可分解为若干子问题，利用动态规划分别解决这些子问题，能够求出中国象棋状态总数的精确解。实验得出中国象棋状态总数约为7.54×10^39.88，过去许多文献描述的中国象棋状态总数是不准确的，远远高估了中国象棋状态总数。基于动态规划的计数方法也可以用于计算其他棋类的空间复杂度，也能够用于寻找空间复杂度较低的残局棋型，为构建中国象棋残局库提供依据。

Rubik's Cube, Chinese Chess ...

Principle of Optimality



https://en.wikipedia.org/wiki/Richard_E._Bellman

Bellman's Principle of Optimality

An optimal policy has the property that whatever the initial state and initial decision are, the remaining decisions must constitute an optimal policy with regard to the state resulting from the first decision

Principle of Optimality, ctd'

The *optimal* path for a multistage decision process is shown in Fig. 3-1(a). Suppose that the first decision (made at a) results in segment a-b with cost

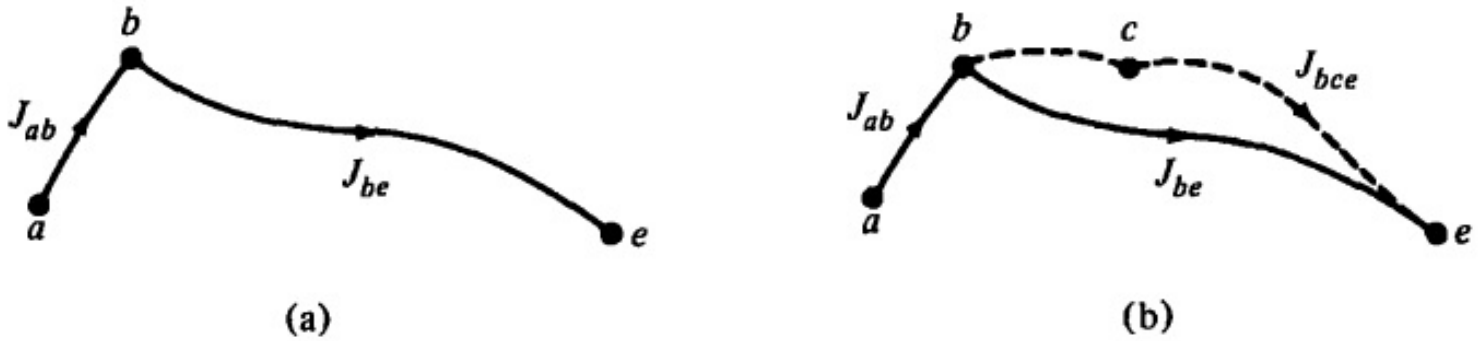


Figure 3-1 (a) Optimal path from a to e. (b) Two possible optimal paths from b to e

J_{ab} and that the remaining decisions yield segment b-e at a cost of J_{be}. The *minimum* cost J_{ae}^* from a to e is therefore

$$J_{ae}^* = J_{ab} + J_{be}. \tag{3.2-1}$$

ASSERTION: If a-b-e is the optimal path from a to e, then b-e is the optimal path from b to e.

DP in Discrete-Time Systems

Computational Procedure

Optimal control (discrete-time version)

$$\min J = h(x_N, N) + \sum_{k=0}^{N-1} g(x_k, u_k, k)$$

subject to $\quad x_{k+1} = a(x_k, u_k, k), \quad k \in \mathbb{Z}_+$

$$x_{k_0} = x_0 \in \mathbb{R}^n \quad \text{fixed}$$

$$M(x_N, N) = 0 \quad \text{terminal state constraint}$$

$$u_k \in \mathcal{U} \subset \mathbb{R}^m \quad \text{control constraint}$$

"Cost-to-go"

$$J^*_{N-k,N}(x_{N-k}) = \min_{u_{N-k} \in \mathcal{U}} \Big\{ g(x_{N-k}, u_{N-k}, N-k) + J^*_{N-k+1,N}\big(a(x_{N-k}, u_{N-k}, N-k)\big) \Big\}$$

for $\quad k = 1, 2, 3, \cdots, N$

Analytical Results - Discrete LQR

Problem of Linear Quadratic Regulator (LQR)

$\boldsymbol{J = x_N^T H x_N + \sum_{k=0}^{N-1} \left(x_k^T Q_k x_k + u_k^T R_k u_k \right)}$ subject to $\boldsymbol{x_{k+1} = A_k x_k + B_k u_k}$

or the time-invariant situation

$\boldsymbol{J = x_N^T H x_N + \sum_{k=0}^{N-1} \left(x_k^T Q x_k + u_k^T R u_k \right)}$ subject to $\boldsymbol{x_{k+1} = A x_k + B u_k}$

Formula/algorithm

- Algorithm
  (i) $\boldsymbol{P_N = H}$
  (ii) $\boldsymbol{F_i = -[R + B^T P_{i+1} B]^{-1} B^T P_{i+1} A}$
  (iii) $\boldsymbol{P_i = Q + F_i^T R F_i + [A + B F_i]^T P_{i+1} [A + B F_i]}$
  or calculate difference Riccati equation, for $\boldsymbol{i = N-1, \cdots, 0}$,

  $$\boldsymbol{P_i = Q + A^T P_{i+1} A - A^T P_{i+1} B [R + B^T P_{i+1} B]^{-1} B^T P_{i+1} A}$$

- Optimal cost: $\boldsymbol{J^* = x_0^T P_0 x_0}$
- Optimal control law: $\boldsymbol{u_k^* = F_k x_k, \quad k = 0, 1, \cdots, N-1}$

Suboptimal Control

Let $J = x_N^T H x_N + \sum_{k=0}^{N-1} \left(x_k^T Q x_k + u_k^T R u_k \right)$ subject to $x_{k+1} = A x_k + B u_k$

Question Compared with the optimal cost: $J^* = x_0^T P_0 x_0$, what is the cost of any different controller $u_k = G_k x_k$?

(i) $S_N = H$

(ii) $S_k = Q + G_k^T R G_k + [A + B G_k]^T S_{k+1} [A + B G_k]$

$k = N-1, N-2, \cdots, 0$

The cost is

$$J = x_0^T S_0 x_0 \quad \text{vs} \quad J^* = x_0^T P_0 x_0$$

Infinite Horizon Problems

Consider LQR problme $J = x_N^T H x_N + \sum_{k=0}^{N-1} \left(x_k^T Q x_k + u_k^T R u_k \right)$ subject to LTI $x_{k+1} = Ax_k + Bu_k$

Assume that H, Q is p.s.d., $Q = C^T C$, and R is p.d.

(i) $P_N = H$
(ii) $F_k = -[R + B^T P_{k+1} B]^{-1} B^T P_{k+1} A$
(iii) $P_k = Q + F_k^T R F_k + [A + BF_k]^T P_{k+1} [A + BF_k]$

Question $H = 0$, $N \to \infty$? If (A, B, C) is controllable and observable,

$$\lim_{k \to \infty} P_k = P_{ss}$$

to be p.d., so that

$$P_{ss} = Q + F_{ss}^T R F_{ss} + [A + BF_{ss}]^T P_{ss} [A + BF_{ss}]$$

$$F_{ss} = -[R + B^T P_{ss} B]^{-1} B^T P_{ss} A$$

or satisfying the algebraic Riccati equation

$$P_{ss} = Q + A^T P_{ss} A - A^T P_{ss} B [R + B^T P_{ss} B]^{-1} B^T P_{ss} A$$

DP in Continuous-Time Systems

HJB Equation

Optimal control (continuous time version)

$$\min J = h(x(t_f), t_f) + \int_t^{t_f} g(x(\tau), u(\tau), \tau) d\tau$$

subject to $\dot{x}(t) = a(x(t), u(t), t)$

$x(t)$ initial condition

$M(x(t_f), t_f) = 0$ terminal state constraint

Hamilton-Jacobi-Bellman Equation (HJB)

$$0 = J_t^*(x(t), t) + \min_{u(t)} \mathcal{H}\big(x(t), u(t), J_x^*(x(t), t), t\big)$$

$$\textbf{or} \quad 0 = J_t^*(x(t), t) + \mathcal{H}\big(x(t), u^*(t), J_x^*(x(t), t), t\big)$$

and $J^*(x(t_f), t_f) = h(x(t_f), t_f)$ **(boundary condition)**

Hamiltonian

$$\mathcal{H}(x, u, J_x^*, t) \triangleq g(x(t), u(t), t) + J_x^{*T}(x(t), t) a(x(t), u(t), t)$$

Continuous LQR

Problem of Linear Quadratic Regulator (LQR)

$$J = x^T(t_f)Hx(t_f) + \int_{t_0}^{t_f}(x(t)^T Q(t)x(t) + u(t)^T R(t)u(t))dt$$

subject to $\dot{x}(t) = A(t)x(t) + B(t)u(t)$

Formula/algorithm

- Differential Riccati equation

$$-\dot{P}(t) = P(t)A + A^T P(t) + Q - P(t)BR^{-1}B^T P(t)$$

with the boundary condition $P(t_f) = H$

- Optimal control law

$$u^* = -\frac{1}{2}R^{-1}B^T J_x^* = -R^{-1}B^T P(t)x$$

- Optimal cost: $J^*(x(t), t) = x^T(t)P(t)x(t)$

Infinite Horizon LQR Problem

Consider LQR problem
$J = x^T(t_f)Hx(t_f) + \int_t^{t_f}(x^TQx + u^TRu)d\tau$ s.t.
LTI $\dot{x} = Ax + Bu$

Assume that H is p.s.d., $Q = C^TC$, and R is p.d.

$-\dot{P}(t) = P(t)A + A^TP(t) + Q - P(t)BR^{-1}B^TP(t)$ with $P(t_f) = H$

Question $t_f \to \infty$?

If (A, B, C) is controllable and observable, then for $H = 0$,

$$\lim_{t\to\infty} P(t) = P_{ss}$$

to be p.d., so that (Algebraic Riccati equation)

$$0 = P_{ss}A + A^TP_{ss} + Q - P_{ss}BR^{-1}B^TP_{ss}$$

cf. Discrete Riccati equation

$$P_{ss} = Q + A^TP_{ss}A - A^TP_{ss}B[R + B^TP_{ss}B]^{-1}B^TP_{ss}A$$

ADP (Numerical Methods)

Motivation Question

Consider infinite horizon LQR problem
$\boldsymbol{J = \int_t^\infty (x^T Q x + u^T R u) d\tau}$ s.t. LTI $\boldsymbol{\dot{x} = Ax + Bu}$

Assume that $\boldsymbol{Q = C^T C}$, $\boldsymbol{R}$ is p.d., and the system triplet $\boldsymbol{(A, B, C)}$ is controllable and observable

Question How to calculate $\boldsymbol{P_{ss}}$?

$$\boldsymbol{0 = P_{ss} A + A^T P_{ss} + Q - P_{ss} B R^{-1} B^T P_{ss}}$$

is nonlinear in $\boldsymbol{P_{ss}}$

Notes on Solving Sylvester Equation (review)

Kronecker Product

The Kronecker product of matrices $A = [a_{ij}] \in \mathbb{R}^{m \times q}$ and $B = [b_{ij}] \in \mathbb{R}^{p \times n}$ is $A \otimes B = [a_{ij}B] \in \mathbb{R}^{mp \times qn}$.

Consider the matrix $X \in \mathbb{R}^{n \times m}$, and let $\mathbf{vec}(X)$ be the column stacking vector of a matrix X. Then

$$\mathbf{vec}(BXA) = (A^T \otimes B)\mathbf{vec}(X)$$

Sylvester Equation

Consider matrices $A \in \mathbb{R}^{m \times m}$, $B \in \mathbb{R}^{n \times n}$, and $Q \in \mathbb{R}^{n \times m}$

Then the Sylvester equation

$$XA - BX = Q \text{ with variable } X \in \mathbb{R}^{n \times m}$$

or rewritten by $(A^T \otimes I_n - I_m \otimes B)\mathbf{vec}(X) = \mathbf{vec}(Q)$ has a unique solution iff A and B have no common eigenvalue

For Proofs, see Appendix A in Huang, J. (2004). Nonlinear Output Regulation: Theory and Applications. SIAM.

Kleinman's Iterative Technique

Consider infinite horizon LQR problem

$\boldsymbol{J = \int_t^\infty (x^T Q x + u^T R u) d\tau}$ s.t. LTI $\boldsymbol{\dot{x} = Ax + Bu}$

Assume that $\boldsymbol{Q = C^T C}$ is p.s.d., $\boldsymbol{R}$ is p.d., and $\boldsymbol{(A, B, C)}$ is controllable and observable.

Algorithm to calculate $\boldsymbol{P_{ss}}$ so that

$$\boldsymbol{0 = P_{ss} A + A^T P_{ss} + C^T C - P_{ss} B R^{-1} B^T P_{ss}}$$

- Select any $\boldsymbol{K_0}$ so that $\boldsymbol{A - BK_0 \triangleq A_0}$ is Hurwitz
- Solve linear equation (**a Sylvester equation**)

$$\boldsymbol{0 = P_0 A_0 + A_0^T P_0 + C^T C + K_0^T R K_0}$$

- For $\boldsymbol{i = 0, 1, 2, 3, \cdots}$, recursion:

$$\boldsymbol{P_i \Longrightarrow K_{i+1} = R^{-1} B^T P_i \Longrightarrow A_{i+1} = A - BK_{i+1} \Longrightarrow P_{i+1}}$$

$\boldsymbol{P_{i+1}}$ is solved from $\boldsymbol{0 = P_{i+1} A_{i+1} + A_{i+1}^T P_{i+1} + C^T C + K_{i+1}^T R K_{i+1}}$

Kleinman's Iterative Technique, ctd'

Consider infinite horizon LQR problem

$\boldsymbol{J = \int_t^\infty (x^T C^T C x + u^T R u) d\tau}$ s.t. LTI $\boldsymbol{\dot{x} = Ax + Bu}$

Assume that $\boldsymbol{Q = C^T C}$, $\boldsymbol{R}$ is p.d., and $\boldsymbol{(A, B, C)}$ is ctr. & obs.

Algorithm

- Start: Select any $\boldsymbol{K_0}$ so that $\boldsymbol{A - BK_0 \triangleq A_0}$ is Hurwitz
- Recursion: for $\boldsymbol{i = 0, 1, 2, 3, \cdots}$

$$\boldsymbol{P_i \Longrightarrow K_{i+1} = R^{-1} B^T P_i \Longrightarrow A_{i+1} = A - BK_{i+1} \Longrightarrow P_{i+1}}$$

- Stop: $\boldsymbol{\|P_{i+1} - P_i\|}$ is small

Convergence

- $\boldsymbol{P_{ss} \leq P_{i+1} \leq P_i}$ for each $\boldsymbol{i = 0, 1, 2, 3, \cdots}$
- $\boldsymbol{\lim_{i \to \infty} P_i = P_{ss}}$ and $\boldsymbol{\lim_{i \to \infty} K_i = K_{ss}}$

Reading: Kleinman, D. (1968). On an iterative technique for Riccati equation computations.

IEEE Transactions on Automatic Control, 13(1), 114–115.

Adaptive Dynamic Programming (ADP)

Consider infinite horizon LQR problem

$\boldsymbol{J = \int_t^\infty (x_\tau^T Q x_\tau + u_\tau^T R u_\tau) d\tau}$ s.t. LTI $\boldsymbol{\dot{x}_t = Ax_t + Bu_t}$

Assume that $\boldsymbol{Q = C^T C}$ is p.s.d., $\boldsymbol{R}$ is p.d., and $\boldsymbol{(A, B, C)}$ is controllable and observable.

Let $\boldsymbol{K_0}$ be such that $\boldsymbol{A - BK_0}$ is Hurwitz.

Cost (quadratic) function

$$\boldsymbol{K_0 \Longrightarrow P_0} \text{ i.e. } \boldsymbol{J(x(t)) = \int_t^\infty x_\tau^T [Q + K_0^T R K_0] x_\tau d\tau = x_t^T P_0 x_t}$$

Split of time interval

$$\underbrace{\boldsymbol{J(x(t))}}_{\boldsymbol{= x_t^T P_0 x_t}} = \int_t^{t+T} \boldsymbol{x_\tau^T [Q + K_0^T R K_0] x_\tau d\tau} + \underbrace{\boldsymbol{J(x(t+T))}}_{\boldsymbol{= x_{t+T}^T P_0 x_{t+T}}}$$

An ADP Algorithm

Consider infinite horizon LQR problem

$J = \int_t^\infty (x_\tau^T Q x_\tau + u_\tau^T R u_\tau) d\tau$ s.t. LTI $\dot{x}_t = Ax_t + Bu_t$

where $Q = C^T C$ is p.s.d., R is p.d., (A, B, C) is ctr. & obs.

Algorithm

- Start: K_0 s.t. $A - BK_0$ is Hurwitz
- Recursion: for $i = 0, 1, 2, 3, \cdots$

$$\underbrace{K_i \Longrightarrow P_i}_{KEY} \Longrightarrow K_{i+1} = R^{-1} B^T P_i$$

$$\underbrace{J(x(t))}_{= x_t^T P_i x_t} = \int_t^{t+T} x_\tau^T [Q + K_i^T R K_i] x_\tau d\tau + \underbrace{J(x(t+T))}_{= x_{t+T}^T P_i x_{t+T}}$$

- Stop: $\|P_{i+1} - P_i\|$ is small

Online Implementation

Consider infinite horizon LQR problem

$\boldsymbol{J = \int_t^\infty (x_\tau^T Q x_\tau + u_\tau^T R u_\tau) d\tau}$ s.t. LTI $\boldsymbol{\dot{x}_t = Ax_t + Bu_t}$

where $\boldsymbol{Q = C^T C}$ is p.s.d., $\boldsymbol{R}$ is p.d., $\boldsymbol{(A, B, C)}$ is ctr. & obs.

Algorithm

- Start: $\boldsymbol{K_0}$ s.t. $\boldsymbol{A - BK_0}$ is Hurwitz
- Recursion: do $\boldsymbol{K_i \Longrightarrow P_i \Longrightarrow K_{i+1} = R^{-1} B^T P_i}$, $\boldsymbol{i = 0, 1, 2, \cdots}$
- Stop: $\boldsymbol{\|P_{i+1} - P_i\|}$ is small

Key Idea: For any $\boldsymbol{T > 0}$, $\boldsymbol{x(t)}$ is measurable in time interval $\boldsymbol{[t, t + T]}$

One may solve the linear equation for $\boldsymbol{P_i}$

$$\underbrace{x_t^T P_i x_t}_{\triangleq\, X_t p_i} = \int_t^{t+T} x_\tau^T [Q + K_i^T R K_i] x_\tau d\tau + \underbrace{x_{t+T}^T P_i x_{t+T}}_{\triangleq\, X_{t+T} p_i} \quad \Longrightarrow \quad P_i$$

Online Implementation, ctd'

Question: How many independent parameters in P_i?

If $x \in \mathbb{R}^n$, then $\mathbf{dim}(p_i) = \frac{1}{2}n(n+1)$ and p_i is NOT $\mathbf{vec}(P_i) \in \mathbb{R}^{n^2}$

More hints

$$X_t p_i = \int_t^{t+T} x_\tau^T [Q + K_i^T R K_i] x_\tau d\tau + X_{t+T} p_i$$

$$\Longrightarrow \quad \underbrace{(X_t - X_{t+T})}_{regression\ vector} \quad \underbrace{p_i}_{unknown} = \underbrace{\int_t^{t+T} x_\tau^T [Q + K_i^T R K_i] x_\tau d\tau}_{measurable}$$

- Denote

  $$\tilde{X}_{t,T} \triangleq X_t - X_{t+T}, \;\; d(X_t, T, K_i) \triangleq \int_t^{t+T} x_\tau^T [Q + K_i^T R K_i] x_\tau d\tau$$

  Both can be measured/evaluated online
- Consider equation

  $$\tilde{X}_{t,T} p_i = d(X_t, T, K_i)$$

Online Implementation, ctd'

- Consider equation: $\tilde{X}_{t,T} p_i = d(X_t, T, K_i)$
- Split the time interval $[t, t+T]$:
  Let $N \geq \frac{1}{2} n(n+1)$ and $l = \frac{1}{N} T$. Then

$$[t, t+l], \quad \underbrace{[t+l, t+2l]}_{length=l}, \quad [t+2l, t+3l], \quad \cdots$$

- Obtain N linear equations

$$\tilde{X}_{t,l} p_i = d(X_t, l, K_i), \quad \tilde{X}_{t+l,l} p_i = d(X_{t+l}, l, K_i), \quad \cdots$$

- Least Squares Estimation Problem $\min_x \|y - Ax\|$

Reading: Vrabie, D., et al (2009). Adaptive optimal control for continuous-time linear systems based on policy iteration. Automatica. 45: 477–484.

Examples (Course Project)

Selected Topic

Modeling, optimal control design, and simulation

- Topic: 2-DOF Helicopter (Quanser Aero Experiment Platform)